

# A degree-two counterexample to the proposed minimal structured determinantal representation

Candidate counterexample to the open question in arXiv:2601.14397

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**Status: candidate full counterexample, likely valid; human review requested.** A concrete symmetric polynomial of bidegree  $(2, 2)$  is zero-free on the *closed* bidisk but has no structured representation from Theorem 2.2 of the source with  $2 \times 2$  blocks. The obstruction is certified in exact rational arithmetic for every possible spectral partition.

## Source question

Baran and Woerdeman prove that a symmetric bidisk-stable polynomial of bidegree  $(n, n)$  has a representation

$$p(z, w) = p(0, 0) \det \left( I - \begin{pmatrix} A_1 & A_2 \\ A_2 & A_1 \end{pmatrix} \begin{pmatrix} zI & 0 \\ 0 & wI \end{pmatrix} \right)$$

with a contractive structured matrix and blocks of size at most  $2n \times 2n$  [1, Theorem 2.2]. Immediately afterward, on source PDF page 4, they ask whether the blocks can always have size  $n \times n$ .

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We will provide the proof of this result in the next section. The results in [13] and [10] suggest that perhaps one can choose  $A_1$  and  $A_2$  to be of size  $n \times n$ , but whether this indeed holds is an open question. We note that [10, Example 2.4] provides an illustration of the above result.

## Counterexample

Define

$$\begin{aligned} f(z, w) = 1 - \frac{111}{200}(z + w) - \frac{69}{250}(z^2 + w^2) + \frac{77}{250}zw \\ + \frac{5451}{20000}(z^2w + zw^2) - \frac{1197}{10000}z^2w^2. \end{aligned} \tag{1}$$

Equivalently, with  $s = z + w$  and  $q = zw$ ,

$$f = 1 - \frac{111}{200}s - \frac{69}{250}s^2 + \frac{43}{50}q + \frac{5451}{20000}sq - \frac{1197}{10000}q^2.$$

## Proof intuition

The stability and the obstruction have separate finite certificates. For stability, regard  $f$  as a quadratic in  $z$  on each circle slice  $|w| = 1$ . The degree-two Schur–Cohn inequalities reduce to three explicit quadratics in  $x = \operatorname{Re} w$ ; their signs on  $[-1, 1]$  follow just from convexity or concavity and endpoint values.

For the obstruction, write  $B = A_1 + A_2$  and  $C = A_1 - A_2$ . A structured contraction is unitarily equivalent to  $B \oplus C$ , so both  $B$  and  $C$  must be contractions, while the determinant becomes

$$\det\left(I - \frac{z+w}{2}(B+C) + zwBC\right).$$

The diagonal polynomial  $f(t, t)$  fixes the four distinct eigenvalues that must be divided between  $B$  and  $C$ . Up to interchange there are three divisions. Coefficient comparison gives one simultaneous-similarity class for each division. Finally, an exact semidefinite dual certificate rules out a common Hilbert-space metric in which both canonical matrices are contractions.

**Theorem 1.** *The polynomial  $f$  in (1) is symmetric, has bidegree  $(2, 2)$ , and has no zero in  $\overline{\mathbb{D}}^2$ . There do not exist  $A_1, A_2 \in M_2(\mathbb{C})$  for which*

$$f(z, w) = \det\left(I - \begin{pmatrix} A_1 & A_2 \\ A_2 & A_1 \end{pmatrix} \begin{pmatrix} zI_2 & 0 \\ 0 & wI_2 \end{pmatrix}\right)$$

and  $\begin{pmatrix} A_1 & A_2 \\ A_2 & A_1 \end{pmatrix}$  is a contraction. Consequently the answer to the source question is negative, already for  $n = 2$  and even under closed-bidisk stability.

*Proof.* Symmetry and exact bidegree are immediate from (1). We prove stability first.

**1. Closed-bidisk stability.** For fixed  $w$ , write

$$f(z, w) = \alpha(w)z^2 + \beta(w)z + \gamma(w),$$

where

$$\begin{aligned}\alpha(w) &= -\frac{69}{250} + \frac{5451}{20000}w - \frac{1197}{10000}w^2, \\ \beta(w) &= -\frac{111}{200} + \frac{77}{250}w + \frac{5451}{20000}w^2, \\ \gamma(w) &= 1 - \frac{111}{200}w - \frac{69}{250}w^2.\end{aligned}$$

For  $|w| = 1$ , the roots of  $f(\cdot, w)$  lie outside  $\overline{\mathbb{D}}$  exactly when the roots of

$$\gamma(w)z^2 + \beta(w)z + \alpha(w)$$

lie in  $\mathbb{D}$ . The quadratic Schur step says that this holds if and only if

$$L := |\gamma|^2 - |\alpha|^2 > 0, \quad |M| < L, \quad M := \bar{\gamma}\beta - \alpha\bar{\beta}. \quad (2)$$

Indeed, subtracting  $\alpha$  times the reversed polynomial from  $\bar{\gamma}$  times the polynomial and dividing by  $z$  gives the linear polynomial  $Lz + M$ .

Put  $x = \operatorname{Re} w$ . Exact expansion on  $|w| = 1$  gives

$$L = \frac{734995123 - 235177572x - 494459520x^2}{400000000}, \quad (3)$$

$$\begin{aligned}L^2 - |M|^2 &= \frac{3}{40000000000000000} (3680x^2 + 1883x - 5801) \\ &\quad \cdot (2807315892240x^2 + 150481096164x - 2964727540151).\end{aligned} \quad (4)$$

The numerator in (3) is concave. Its values at  $-1$  and  $1$  are respectively

$$475713175 \quad \text{and} \quad 5358031,$$

so  $L > 0$  on  $[-1, 1]$ . Both quadratic factors in (4) are convex, and their endpoint values are

$$(-4004, -238), \quad (-307892744075, -6930551747).$$

A convex function on an interval is bounded above by the larger endpoint value; hence both factors are strictly negative throughout  $[-1, 1]$ . Thus (4) is positive, so (2) holds. Therefore

$$f(z, w) \neq 0 \quad \text{whenever } |w| = 1, |z| \leq 1. \quad (5)$$

By symmetry,  $f(z, w) \neq 0$  also whenever  $|z| = 1$  and  $|w| \leq 1$ . For fixed  $w \in \overline{\mathbb{D}}$ , the number of zeros of  $f(\cdot, w)$  in  $\mathbb{D}$  is consequently constant as  $w$  moves to 0. At  $w = 0$ ,

$$f(z, 0) = 1 - \frac{111}{200}z - \frac{69}{250}z^2, \quad \frac{111}{200} + \frac{69}{250} = \frac{831}{1000} < 1,$$

so Rouché's theorem gives no zero in  $\overline{\mathbb{D}}$ . Together with (5), this proves that  $f$  is zero-free on  $\overline{\mathbb{D}}^2$ .

**2. Reduction to three simultaneous-similarity classes.** Suppose that a representation with  $2 \times 2$  blocks exists. Set

$$B = A_1 + A_2, \quad C = A_1 - A_2.$$

The standard Hadamard unitary diagonalizes the structured block matrix to  $B \oplus C$ . Hence  $B$  and  $C$  are contractions. The determinant identity from the source calculation becomes

$$f(z, w) = \det\left(I - \frac{z+w}{2}(B+C) + zwBC\right). \quad (6)$$

On the diagonal,

$$f(t, t) = -\frac{(3t-10)(3t-4)(7t+10)(19t-25)}{10000}. \quad (7)$$

But (6) at  $z = w = t$  factors as  $\det(I - tB) \det(I - tC)$ . Thus the spectra of  $B$  and  $C$  form a  $2 + 2$  partition of

$$\left\{ \frac{3}{10}, \frac{3}{4}, -\frac{7}{10}, \frac{19}{25} \right\}. \quad (8)$$

All four numbers are distinct. Diagonalize  $B = \text{diag}(b_1, b_2)$  and write  $C = \begin{pmatrix} u & x \\ y & v \end{pmatrix}$ . The trace and determinant of  $C$  are fixed by its two assigned eigenvalues. Moreover, comparison of the  $z^2$  coefficient in (6) gives

$$-\frac{69}{250} = \frac{1}{4}(b_1b_2 + b_1v + b_2u + \det C). \quad (9)$$

Hence  $u, v$  and  $xy = uv - \det C$  are fixed. In every case  $xy \neq 0$ , and a remaining diagonal similarity makes  $x = 1$ . Up to interchanging  $B$  and  $C$ , the three possible canonical pairs are

| $\sigma(B)$                       | $B$                                                                   | $C$                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $\{\frac{3}{10}, -\frac{7}{10}\}$ | $\begin{pmatrix} \frac{3}{10} & 0 \\ 0 & -\frac{7}{10} \end{pmatrix}$ | $\begin{pmatrix} \frac{1917}{1000} & 1 \\ -\frac{1350219}{1000000} & -\frac{407}{1000} \end{pmatrix}$ |
| $\{\frac{3}{10}, \frac{3}{4}\}$   | $\begin{pmatrix} \frac{3}{10} & 0 \\ 0 & \frac{3}{4} \end{pmatrix}$   | $\begin{pmatrix} -\frac{163}{90} & 1 \\ -\frac{1157}{405} & \frac{421}{225} \end{pmatrix}$            |
| $\{\frac{3}{10}, \frac{19}{25}\}$ | $\begin{pmatrix} \frac{3}{10} & 0 \\ 0 & \frac{19}{25} \end{pmatrix}$ | $\begin{pmatrix} -\frac{411}{230} & 1 \\ -\frac{5835}{2116} & \frac{169}{92} \end{pmatrix}$           |

The complementary spectral partitions merely interchange  $B$  and  $C$ ; this sends  $A_2$  to  $-A_2$  and conjugates the original structured block matrix by  $\text{diag}(I, -I)$ , leaving its determinant and contractivity unchanged.

**3. Exact common-metric obstruction.** If a canonical pair is simultaneously similar to two contractions, then some Hermitian  $H > 0$  satisfies

$$B^*HB \leq H, \quad C^*HC \leq H. \quad (10)$$

For positive semidefinite matrices  $X, Y$ , put

$$Z = X - BXB^* + Y - CYC^*.$$

If  $Z < 0$ , then (10) is impossible, because

$$\begin{aligned} 0 &\leq \operatorname{tr}(X(H - B^*HB)) + \operatorname{tr}(Y(H - C^*HC)) \\ &= \operatorname{tr}(ZH) < 0. \end{aligned} \tag{11}$$

For the three rows above, take respectively the following exact integer matrices:

$$\begin{aligned} X_1 &= \begin{pmatrix} 33335537 & -37673850 \\ -37673850 & 42576768 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 23136281 & -4691714 \\ -4691714 & 951414 \end{pmatrix}, \\ X_2 &= \begin{pmatrix} 13149208 & 32661562 \\ 32661562 & 81130750 \end{pmatrix}, & Y_2 &= \begin{pmatrix} 595772 & -1744823 \\ -1744823 & 5124270 \end{pmatrix}, \\ X_3 &= \begin{pmatrix} 13137533 & 32657413 \\ 32657413 & 81181395 \end{pmatrix}, & Y_3 &= \begin{pmatrix} 661546 & -1808353 \\ -1808353 & 5019526 \end{pmatrix}. \end{aligned}$$

Their exact principal-minor checks are:

| $i$ | $\det X_i$  | $\det Y_i$  | $(Z_i)_{11}$           | $\det Z_i$                              |
|-----|-------------|-------------|------------------------|-----------------------------------------|
| 1   | 451181916   | 1393538     | $-14515027401809/10^6$ | $21068602807073965497383976353/10^{14}$ |
| 2   | 27474666156 | 8489285111  | $-339009736/405$       | $127701193764655494359/182250000$       |
| 3   | 16631945966 | 50506774587 | $-51745670959/52900$   | $267815854970301230200709/279841000000$ |

Each  $X_i, Y_i$  has positive upper-left entry and positive determinant, hence is positive definite. Each  $Z_i$  has negative upper-left entry and positive determinant, hence is negative definite. Equation (11) excludes all three classes and their complements. This contradicts the assumed representation and completes the proof.  $\square$

## Verification and reproducibility

The exact checker in `code/verify_counterexample.py` uses symbolic rational arithmetic. It independently checks the displayed determinant, equations (3)–(4), all six spectral assignments (three representatives plus complements), positivity of every  $X_i, Y_i$ , and negativity of every  $Z_i$ . It was run with

```
conda run --no-capture-output -n sandbox
python code/verify_counterexample.py
```

and returned `all exact checks passed`. No numerical approximation is used in the proof.

The most important human-review points are the Schur–Cohn transition from the side boundary to the full bidisk and the completeness of the canonical-pair classification after (9).

## Novelty check and limitations

On 11 August 2026, bounded arXiv searches used the exact source title and phrase, together with the terms *symmetric stable polynomial*, *bidisk*, *structured determinantal representation*, *Kummert*, and  $n \times n$ . The search recovered the source paper and earlier general determinantal-realization papers, but no later answer or matching counterexample. The source was posted in January 2026, so the search window was necessarily short. Novelty is therefore plausible but not established beyond this bounded search.

The theorem only disproves the proposed  $n \times n$  structured size bound. It does not challenge the source theorem’s larger  $2n \times 2n$  bound or unstructured determinantal representations. The packet should be checked by an expert before public use.

## References

- [1] R. Baran and H. J. Woerdeman, *Symmetric Schur-class functions on the bidisk and Schur-class functions on the symmetrized bidisk*, arXiv:2601.14397v1 (2026), Theorem 2.2 and p. 4.